Practice Problem.

Consider a certain harbor where the tides cause the ocean's surface to rise and fall a distance D from the highest to the lowest level in simple harmonic motion (SHM). If the period of this SHM is one full day, determine the time taken (in hours) for the water to fall a distance $\frac{D}{4}$ from its highest level.

Solution:

$\Rightarrow$ highest to lowest = D or $2x_m$

$\Rightarrow$ period $\Rightarrow T = \frac{2\pi}{\omega} = 24$ hrs

$\therefore$ the amplitude $\Rightarrow x_m = \frac{D}{2}$

$\omega = \frac{2\pi}{24}$ rad/h $= 0.2618$ rad/h

Sim: $x(t) = x_m \cos(\omega t + \phi)$

Since our clock starts at the tides highest point: $x(0) = x_m \Rightarrow \phi = 0$.

$\Rightarrow x(t) = x_m \cos(\omega t + \phi)$

$\frac{D}{4} = \frac{D}{2} \cos(0.2618t + 0)$

NEXT PAGE →

$$\frac{D}{4} = \frac{D}{2}\cos(0.2618t + 0)$$

$$\cos(0.2618t) = 0.5$$

$$\Rightarrow \quad t = \frac{\cos^{-1}(0.5)}{0.2618} = 4 \text{ hr} \quad \#$$